Erfan Ziad

Ph.D. Candidate | Deep Learning, Continual Learning & Generative AI
(206) 602-4501 | erfanziad78@gmail.com | linkedin.com/in/erfan-ziad | github.com/erfanziad | Tempe, AZ

Summary

Ph.D. candidate in Industrial Engineering at Arizona State University (GPA: 4.0/4.00) with 4 years of research experience in deep learning, self-supervised and continual learning, and generative AI, developed through collaborations with NIST and Honeywell Aerospace. First author on 7 published or accepted papers. Core research problems center on learning under distribution shift, data scarcity, and non-stationary environments – directly relevant to continual learning. Available for Summer 2027; graduating May 2027.

Technical Skills

Continual and Self-Supervised Learning: BYOL, distribution shift, domain adaptation, transfer learning, few-shot learning, knowledge-constrained learning, catastrophic forgetting
Generative Models: Diffusion models (DDPM, conditional), GANs (WGAN-GP), synthetic data generation, fidelity-latency benchmarking
Architectures: Vision Transformers (SWIN, ViT), CNNs, U-Net, ResNet, LSTMs, multimodal fusion, attention mechanisms
Theory: Convex optimization, linear algebra, probability theory, statistical learning theory, stochastic processes
Programming: Python (PyTorch, NumPy, Pandas, Scikit-learn, OpenCV), Julia, C (basic), R (basic)
Tools: Git, HuggingFace, HPC4Health, VS Code, Docker (basic)

Research Experience

Graduate Research Assistant – Deep Learning & Self-Supervised Learning 05/2023 – Present
Arizona State University | NIST Collaboration (Grant 70NANB22H183) *Tempe, AZ*

- Developed a BYOL-based self-supervised continual pre-training pipeline in PyTorch on sequential unlabeled image batches – then fine-tuned for multi-label dense defect segmentation under distribution shift – achieving mIoU = 0.71 while reducing labeled data requirements by over 60% versus fully supervised baselines, directly addressing the core data-efficiency problem in continual learning.
- Designed and benchmarked conditional WGAN-GP and diffusion model architectures for domain-specific synthetic data generation in PyTorch, evaluating generation fidelity (FID) and downstream task utility across distribution shift conditions – demonstrating that diffusion models achieve higher fidelity while GANs offer lower latency, quantified tradeoff useful for continual learning data augmentation. Accepted ASME MSEC 2026.
- Built a knowledge-constrained deep clustering algorithm in PyTorch for unsupervised anomaly detection under zero labeled examples, enabling detection of novel and previously unseen failure modes – directly relevant to open-world and class-incremental continual learning. Published IEEE CASE 2024, 26 citations.
- Developed a multimodal Transformer fusing heterogeneous in-situ sensor streams (camera images and photodetector time-series) using cross-modal attention in PyTorch, achieving $R^2 = 0.95$ on voxel-level quality prediction – submitted Elsevier Additive Manufacturing 2026.
- Collaborated with NIST researchers in a multi-institutional program; delivered 7 peer-reviewed publications as first author across ASME, IEEE, and Elsevier venues.

Research Intern – AI under Distribution Shift in Clinical Environments 10/2026 – 12/2026
University Health Network (Princess Margaret Cancer Centre) *Toronto, Canada (Remote)*

- Developing machine learning pipelines for the AIM2Reduce project on HPC4Health infrastructure, applying deep learning models to heterogeneous clinical datasets with significant patient population distribution shift across cohorts – a real-world continual learning deployment challenge.
- Built LLM-based pipelines using Ollama (llama3.1:8b) for adverse event extraction and CARG toxicity score estimation from unstructured clinical notes – demonstrating adaptation of pre-trained models to new domain data without full retraining.

Research Assistant – Prognostics under Non-Stationary Distributions 2024 – Present
Arizona State University | Honeywell Aerospace Collaboration *Tempe, AZ*

- Designed a three-stage anomaly detection and prognostics pipeline for Honeywell Aerospace analyzing 38 APUs across 5,521 flights and 210 fault events – applied Mann-Kendall drift tests with prewhitening to detect non-stationary parameter shifts in APU sensor streams (EGT, shaft speed, oil temperature), providing empirical evidence that standard i.i.d. assumptions fail in long-horizon aerospace monitoring, motivating continual learning approaches.
- Built multi-tab Plotly Dash health monitoring dashboard with composite health scoring and fleet-percentile thresholds, enabling real-time detection of distributional drift in APU operating parameters.

Instructor of Record – IEE 474: Quality Control 01/2025 – 05/2025
Arizona State University *Tempe, AZ*

- Designed and deployed a RAG-based LLM grading agent using the Claude API and Python, automating assignment evaluation for 40+ students while maintaining rubric consistency – reducing grading turnaround from days to hours.

Selected Projects

BYOL Self-Supervised Continual Pre-training | *PyTorch, Python* 2025

- Implemented BYOL from scratch for sequential self-supervised pre-training on unlabeled image batches under distribution shift, then fine-tuned for multi-label dense segmentation – achieved mIoU = 0.71 with 60%+ reduction in labeled data versus supervised baselines. Published IEEE CASE 2025. github

Generative Benchmark: GAN vs. Diffusion under Distribution Shift | *PyTorch* 2026

- Built automated benchmarking pipeline comparing conditional WGAN-GP and diffusion architectures for synthetic data generation – evaluated fidelity (FID), latency, and downstream task utility across shift conditions. Accepted ASME MSEC 2026. github

Knowledge-Constrained Clustering for Novel Class Detection | *PyTorch* 2024

- Designed unsupervised deep clustering with domain constraints to detect novel and unseen failure classes without labeled examples – published IEEE CASE 2024, 26 citations. Directly relevant to open-world continual learning. github

Multimodal Cross-Modal Attention Transformer | *PyTorch* 2025

- Built Transformer fusing image and time-series sensor streams via cross-modal attention, resolving geometric misalignment across modalities – achieved $R^2 = 0.95$ on held-out quality prediction. github

Non-Stationary Drift Detection in Aerospace Sensor Streams | *Python, Pandas* 2024–Present

- Applied Mann-Kendall tests with prewhitening to detect distributional drift in APU sensor streams across 38-aircraft fleet (5,521 flights) – provided empirical evidence that sensor distributions shift significantly over the aircraft lifecycle, motivating continual model updating.

SWIN Transformer – Cross-Domain Transfer Learning | *PyTorch* 2024

- Fine-tuned SWIN Vision Transformer on ChestX-ray14 for multi-label classification; benchmarked ResNet18/50 on MedMNIST 2D/3D – studying how pre-trained representations transfer across domain boundaries under limited target-domain data.

Publications

Additive Manufacturing, Elsevier (submitted) | **ASME MSEC 2026** (accepted) 2026
ASME JMSE Vol. 148 | **IEEE CASE 2025** | **IEEE CASE 2024** | **ASME MSEC 2024** 2024–2026

Education

Arizona State University Tempe, AZ
Ph.D. in Industrial Engineering *05/2025 – 05/2027 (expected)*

- GPA: 4.0/4.00 | Advisor: Dr. Feng Ju | NIST Collaboration

Arizona State University Tempe, AZ
M.S. in Industrial Engineering (completed May 2025) *01/2023 – 05/2025*

- GPA: 4.0/4.00

Sharif University of Technology Tehran, Iran
B.Sc. in Industrial Engineering *09/2018 – 08/2022*

- GPA: 3.73/4.00

Honors and Awards

2nd Place – IEEE CASE 2025 Hackathon (generative AI, international competition) 2025

Harold and Lucille Dunn Memorial Engineering Scholarship 2024–2025

Fulton Schools of Engineering Fellowship – Exceptional Academic Performance 2024

ASU Alumni Association Scholarship 2024

Ranked 377th of 144,000+ – Nationwide University Entrance Exam (Iran) 2018